ChatHealthy.ai

ARCH-001 Refactor Design

Consolidated Design Document

Iterations 1, 3, and 5 merged into final specification

Date: 2026-03-30

Claude Opus 4.6 (engineer) + GPT-5.3 (architect), 5 iterations, 2026-03-27 through 2026-03-30

1. Executive Summary

Modular monolith refactor of FindCare backend from a 1,323-line main.py monolith to a facade-oriented domain architecture. Two business domains (FindCare, EvaluateCareQuality) with explicit service classes per UAT feature. HuggingFace reduced to entrypoint + adapter only. Strangler Fig migration in 7 phases, zero downtime, deployable at every phase. Pydantic-enforced tool boundaries. Full Perplexity audit response covering all 18 findings.

2. Top-Level Architecture

Pattern: Modular Monolith (Facade-Oriented Domains)

Dependency Direction: Host -> Application -> Domain -> Infrastructure (no upward imports)

Layers

- Host Adapter (HuggingFace only -- app.py + HostChatAdapter)
- Application Layer (ChatOrchestrator + Facades + ToolRouter)
- Domain Layer (Business Services per UAT feature)
- Infrastructure Layer (MongoDB, External APIs, Embeddings, Config)

3. Business Components

3.1 FindCareFacade - Core provider discovery and specialty identification

Service Class	File	UAT Feature	Responsibility
ProviderSearchService	domain/find_care/provider_search_service.py	Provider Search (UAT Feature 1)	Vector search, query expansion, FIPS county mapping, re formatting
SpecialtyService	domain/find_care/specialty_service.py	Specialty Identification (UAT Feature 2)	NLP taxonomy mapping, specialty code lookup

Facade Methods

- search_providers(criteria: ProviderSearchInput) -> ProviderSearchOutput
- identify_specialty(input: SpecialtyInput) -> SpecialtyOutput
- get_provider_location(npi: str) -> dict

3.2 EvaluateCareFacade - Clinical trials search and provider quality evaluation

Service Class	File	UAT Feature	Responsibility

ClinicalTrialsService	domain/evaluate_care_quality/clinical_trials_service.py	Clinical Trials Search (UAT Feature 3)	ClinicalTrials.gov API integration, result filtering
ProviderDetailService	domain/evaluate_care_quality/provider_detail_service.py	Provider Detail (UAT Feature 8)	NPI lookup, external provider data aggregation

Facade Methods

- search_clinical_trials(input: ClinicalTrialsInput) -> ClinicalTrialsOutput
- get_provider_details(npi: str) -> ProviderDetailOutput

Cross-Domain Calls

- EvaluateCareFacade calls FindCareFacade.get_provider_location() for travel calculations

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4. Shared Infrastructure Services

Service Class	File	UAT Feature	Responsibility
SafetyService	domain/shared/safety/safety_service.py	Safety Filter (UAT Feature 5)	IP locking, session safety state, pre-tool gating
EmergencyDetector	domain/shared/safety/emergency_detector.py	Emergency Detection (F-09 fix)	Config-driven keyword + embedding similarity detection
ConsentService	domain/shared/consent/consent_service.py	Consent Framework (UAT Feature 7)	Full two-tier HIPAA consent workflow -- see Section 4.1 Consent F Detail
LeadService	domain/shared/lead_capture/lead_service.py	Lead Capture (UAT Feature 6)	Contact record creation, consent integration
AboutService	domain/shared/content/about_service.py	About ChatHealthy (UAT Feature 4)	Static content loading for about intents
UnknownQuestionService	domain/shared/unknowns/unknown_question_service.py	Unanswerable Question Handling (UAT Feature 12)	De-identified logging of unanswerable questions
URLGuardian	domain/shared/url/url_guardian.py	URL Validation (UAT Feature 9)	Dual-mode URL validation (tool results + LLM text)

4.1 Consent Framework -- Full Workflow Design

HIPAA-aligned two-tier consent framework controlling what user data is stored and how. Consent is a full-time workflow -- not a flag. It governs the entire data lifecycle of a user session.

Trigger: Every 5th user message, system prompt instructs Claude to ask: 'Would you like someone from ChatHealthy.AI to follow up with you personally?'

Two-Tier Consent Flow

Tier 1: Verbatim Consent

Prompt: "May we save a verbatim transcript of this conversation with your contact details?"

If YES:

- consent_verbatim: True
- stored: Full chat_history (PII intact), email, name, notes, reason_for_contact
- evidence: Consent exchange itself is included in the stored transcript

If NO: Proceed to Tier 2

Tier 2: De-Identified Summary Consent

Prompt: "May we save a de-identified summary of this conversation instead?"

If YES:

- consent_verbatim: False
- consent_summary: True
- stored: LLM-generated summary passed through deIdentify() -- no PII, no chat_history
- process: Claude summarizes -> deIdentify() scrubs PII -> stored in notes field

If NO:

- consent_verbatim: False
- consent_summary: False
- stored: Contact fields only (email, name). No history, no summary.

Data Streams

Stream	Consent Required	Treatment	Collection
Questions (unanswerable)	False	Always de-identified before storage. No PII. No consent needed.	AboutUs.AboutSkip
Contact / Lead Capture	True	Two-tier consent governs what is stored. User chooses verbatim or summary.	AboutUs.lead

ConsentService Methods

- request_consent(session_id) -> ConsentState
- record_verbatim_consent(session_id, chat_history, contact_info) -> LeadRecord
- record_summary_consent(session_id, summary, contact_info) -> LeadRecord
- record_no_consent(session_id, contact_info) -> LeadRecord
- de_identify(text: str) -> str

Integration Points

- ChatOrchestrator: triggers consent prompt every 5th message
- LeadService: delegates to ConsentService for data handling
- System Prompt: RULE 7 defines the consent UX flow
- deIdentify(): shared utility used by ConsentService and UnknownQuestionService

Regulatory Notes

- Consent exchange itself must be preserved as evidence (included in verbatim transcript)
- De-identification must be applied BEFORE storage, not after
- Emergency records (BTC-003) must NEVER be deleted regardless of consent state
- Test data tagged with testdata=true for filtering

Lead Record Schema

- email: string
- name: string
- notes: string (summary if consent_summary=true, else empty)
- reason_for_contact: string
- consent_verbatim: boolean
- consent_summary: boolean or null
- chat_history: array (present ONLY if consent_verbatim=true)
- datetime: ISO string
- testdata: boolean

5. HuggingFace Surface Contract

Allowed Files

- app.py
- requirements.txt
- static/

Responsibilities

- HTTP routing (FastAPI app creation, CORS, endpoints)
- Request/response translation
- Stream relay
- NO business logic

Adapter Class: HostChatAdapter

Adapter File: adapter/host_chat_adapter.py

Entrypoint Final State

main.py: DELETED

Replacement: app.py at Code/ConversationalUX/FindCareChat/app.py

- FastAPI app initialization
- Route definitions (/chat, /health, /welcome)
- Delegates ALL logic to HostChatAdapter

6. Facade Design

FindCareFacade (application/facades/find_care_facade.py)

Methods:

- search_providers(criteria: ProviderSearchInput) -> ProviderSearchOutput
- identify_specialty(input: SpecialtyInput) -> SpecialtyOutput
- get_provider_location(npi: str) -> dict

Delegates to: ProviderSearchService, SpecialtyService

EvaluateCareFacade (application/facades/evaluate_care_facade.py)

Methods:

- search_clinical_trials(input: ClinicalTrialsInput) -> ClinicalTrialsOutput
- get_provider_details(npi: str) -> ProviderDetailOutput

Delegates to: ClinicalTrialsService, ProviderDetailService

Cross-calls: FindCareFacade.get_provider_location()

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7. Tool Registry (ToolRouter)

Class: ToolRouter | File: application/tool_router.py | Pattern: Pydantic-validated allowlist dispatch

Tool Name	Pydantic Model	Handler
find_providers	ProviderSearchInput	FindCareFacade.search_providers
find_specialty_codes	SpecialtyInput	FindCareFacade.identify_specialty
search_clinical_trials	ClinicalTrialsInput	EvaluateCareFacade.search_clinical_trials
lookup_provider_external	ProviderDetailInput	EvaluateCareFacade.get_provider_details
get_skip_snow_context	None	AboutService.get_skip_snow_context
get_chathealthy_context	None	AboutService.get_chathealthy_context
record_user_details	LeadInput	LeadService.record_user_details
record_unknown_question	UnknownInput	UnknownQuestionService.record
get_provider_location	LocationInput	FindCareFacade.get_provider_location

Enforcement Rules

- Reject unknown tools (F-05 fix -- eliminates globals().get)
- Validate all inputs via Pydantic models
- Validate outputs before returning to LLM
- Server-side enforcement (F-17 fix)

8. Repository File Tree

```
Code/ConversationalUX/FindCareChat//
  app.py: HuggingFace entrypoint only (~30 lines)
  adapter//
    host_chat_adapter.py: Translates HTTP <-> ChatOrchestrator
  application//
    chat_orchestrator.py: LLM loop, system prompt, safety gating
    tool_router.py: Pydantic-validated tool dispatch (allowlist enforced)
    session_manager.py: Chat session + UX state + timer
    facades//
      find_care_facade.py: Entry point for all FindCare capabilities
      evaluate_care_facade.py: Entry point for all EvaluateCareQuality capabilities
    tool_models//
      provider_search_models.py: Pydantic input/output for provider search
      clinical_trials_models.py: Pydantic input/output for clinical trials
      shared_models.py: Common tool schemas (LeadInput, UnknownInput, etc.)
  domain//
    find_care//
      provider_search_service.py: Feature 1 -- vector search, query expansion, formatting
      specialty_service.py: Feature 2 -- NLP taxonomy mapping
    evaluate_care_quality//
      clinical_trials_service.py: Feature 3 -- ClinicalTrials.gov integration
      provider_detail_service.py: Feature 8 -- NPI lookup, external data
    shared//
      safety//
        safety_service.py: Feature 5 -- IP locking, session safety
        emergency_detector.py: F-09 -- config-driven emergency detection
      consent//
        consent_service.py: Feature 7 -- two-tier HIPAA consent
      lead_capture//
        lead_service.py: Feature 6 -- contact record creation
      content//
        about_service.py: Feature 4 -- about intents
      unknowns//
        unknown_question_service.py: Feature 12 -- unanswerable question logging
      url//
        url_guardian.py: Feature 9 -- dual-mode URL validation (IMPLEMENTED)
  infrastructure//
    mongo//
      provider_repository.py: Read-only provider queries
    external_apis//
```



```
      npi_client.py: NPI registry lookups
      clinical_trials_client.py: ClinicalTrials.gov API
      google_routes_client.py: Travel time calculations
    embeddings//
      embedding_client.py: Vector search client
    config//
      config_loader.py: Loads brain artifacts + environment variables
  frontend//
    src//
      components//
        ChatWindow.tsx
        MessageBubble.tsx
        EmergencyBanner.tsx
        TypingIndicator.tsx
      api//
        chat_client.ts
      state//
        chat_store.ts
      rendering//
        markdown_renderer.tsx
      styles//
        chat.css
Code/DataPipelines//
  status: No structural change (GOV-005 preserved)
  security_patch: Add APIKeyMiddleware to status endpoint (F-06)
Code/Shared//
  purpose: Cross-application infrastructure only (NO business logic)
  logging/: Central structured logging
  config/: Shared env loader
  contracts/: Optional shared DTOs (non-domain)
Website//
  status: No change -- remains Cloudflare static (GOV-005)
  note: Remove any leaked architecture docs (F-13)
```

9. Migration Plan (7 Phases)

Strategy: Strangler Fig pattern inside same codebase (incremental extraction, zero downtime)

Phase	Name	Build	Risk	Key Changes
1	Introduce Skeleton Architecture	B-187	LOW	Create folder structure (application/, domain/, infrastructure/, adapter/); Add ChatOrchestrator (empty pass-throu Add ToolRouter with static mapping (still calls existing main.py functions) (+1 more)
2	Extract FindCare -- Provider Search	B-190	LOW	Move find_providers() -> ProviderSearchService.search(); Introduce FindCareFacade.search_providers(); Upc ToolRouter to call facade instead of main.py
3	Extract FindCare -- Specialty Service	B-191	LOW-MEDIUM	Move find_specialty_codes() to SpecialtyService; Wire through FindCareFacade; Remove corresponding logic f main.py
4	Extract EvaluateCareQuality Domain	B-193	MEDIUM	Move ClinicalTrialsService; Move ProviderDetailService; Introduce EvaluateCareFacade (+1 more)
5	Extract Shared Services	B-194	MEDIUM	SafetyService + EmergencyDetector; ConsentService; LeadService (+3 more)
6	Introduce Pydantic Tool Models	B-195	MEDIUM	Define all tool input/output schemas; Wrap ToolRouter validation layer; Gradually enforce typing per tool
7	Delete main.py Monolith	B-196	LOW (if prior phases complete)	Remove remaining functions from main.py; Ensure all calls go through orchestrator + services; app.py HostChatAdapter is the sole entry point (+1 more)

Safety Mechanisms

- Each phase preserves working state
- Feature-by-feature migration (no big bang)
- Fallback: old functions remain callable until replaced
- Full regression test after Phase 7 (B-197)

10. Testing Strategy

Test Structure

```
tests/application/
    test_chat_orchestrator.py
tests/domain/find_care/
    test_provider_search_service.py
    test_specialty_service.py
tests/domain/evaluate_care_quality/
    test_clinical_trials_service.py
    test_provider_detail_service.py
tests/domain/shared/
    test_safety_service.py
    test_emergency_detector.py
    test_consent_service.py
    test_lead_service.py
    test_unknown_question_service.py
tests/infrastructure/
    test_provider_repository.py
    test_external_clients.py
```

Approach

unit_tests: All domain services with mocked dependencies

integration_tests:

- ToolRouter + Facades
- ChatOrchestrator with fake LLM responses

e2e_tests: Real systems (MongoDB, Anthropic) -- no mocks per Boss directive

regression:

- Capture baseline outputs from main.py before refactor
- Replay test prompts and compare outputs

existing_tests: Retained initially, then migrated per service

11. Pydantic Strategy

Scope: All tool inputs and outputs validated via Pydantic models

Location: application/tool_models/

Models

- ProviderSearchInput / ProviderSearchOutput
- SpecialtyInput / SpecialtyOutput
- ClinicalTrialsInput / ClinicalTrialsOutput
- ProviderDetailInput / ProviderDetailOutput
- LeadInput / LeadOutput
- UnknownInput / UnknownOutput
- LocationInput

Enforcement

- ToolRouter validates input before calling handler
- ToolRouter validates output before returning to LLM
- Schema mismatches raise ValidationError (not silently ignored)

12. Perplexity Audit Response (All 18 Findings)

Audit Source: Perplexity external audit v1 (2026-03-28)

Total Findings: 18

Resolved: 8 | Deferred: 6 | Acknowledged: 4 | Strengths: 1

ID	Title	Status	Response
F-01	Architecture alignment	CONFIRMED	Three-application boundary rules preserved. GOV-005 enforced throughout refactor.
F-02	DataPipelines independence	STRENGTH	DataPipelines remains independent. No structural changes. GOV-005 alignment confirmed.
F-03	MongoDB connection pooling	ACKNOWLEDGED	Connection pooling design deferred to hardening phase (QA/beta). Not in ARCH-001 scope.
F-04	Embedding model consistency	ACKNOWLEDGED	Embedding client centralized in infrastructure layer. Model version pinned in config.
F-05	Tool dispatch via globals().get	RESOLVED	ToolRouter allowlist replaces globals() dispatch. Phase 1 (B-187).
F-06	Pipeline status endpoint unprotected	DEFERRED	Add APIKeyMiddleware to DataPipelines status endpoint. Pipeline concern, not chat refactor. Deferred to next sprint.
F-07	Monolithic main.py	RESOLVED	main.py eliminated in Phase 7 (B-196). Replaced by modular domain architecture.
F-08	All-state coverage gaps	DEFERRED	Pipeline data run, not code change. Deferred to next sprint.
F-09	Emergency keyword list too narrow	RESOLVED	EmergencyDetector with config-driven keywords from brain/machine_artifacts/emergency_keywords.json. Phase 5 + B-189.
F-10	Provider search result cap too low	RESOLVED	Configurable limit in ProviderSearchService (default 25). B-192.
F-11	FIPS hack resolution	DEFERRED	Pipeline re-run required. Deferred to next sprint.
F-12	DeepSeek privacy disclosure	DEFERRED	Legal/copy concern. Deferred to next sprint.
F-13	Architecture docs exposure	DEFERRED	Boss decision required on access gate. Website removes sensitive docs.
F-14	Session management	RESOLVED	SessionManager class in application layer. Chat state + timer centralized.
F-15	Error handling consistency	RESOLVED	Domain services return typed results. Infrastructure layer handles retries. Application layer catches and formats errors.
F-16	Brain lock TTL	ACKNOWLEDGED	BrainLockManager with TTL design documented. Implementation deferred to hardening phase.
F-17	Server-side tool enforcement	RESOLVED	ToolRouter + Pydantic validation enforces server-side. No tool can be called without being in the allowlist.
F-18	Logging and observability	ACKNOWLEDGED	Central structured logging in Code/Shared/logging/. Full observability deferred to hardening phase per project_hardening_phase.md.

13. Sprint Schedule (Tue-Sun, B-187 through B-201)

Sprint Start: 2026-03-31

Sprint Goal: Refactor architecture (ARCH-001), maps, current functionality enhancements

Day	Theme	Build	Task	Risk	Deploy
2026-03-31 (Tuesday)	Skeleton + Security Fix + Maps	B-187	ARCH-001 Phase 1: Skeleton Architecture + ToolRouter	LOW	dev
2026-03-31 (Tuesday)	Skeleton + Security Fix + Maps	B-188	Maps: Provider search results on map	LOW-MEDIUM	dev
2026-03-31 (Tuesday)	Skeleton + Security Fix + Maps	B-189	F-09: Expand emergency keyword list	LOW	dev
2026-04-01 (Wednesday)	Extract FindCare Domain	B-190	ARCH-001 Phase 2: Extract ProviderSearchService	LOW	dev
2026-04-01 (Wednesday)	Extract FindCare Domain	B-191	ARCH-001 Phase 3: Extract SpecialtyService	LOW-MEDIUM	dev
2026-04-01 (Wednesday)	Extract FindCare Domain	B-192	F-10: Raise provider search result cap to 25	LOW	dev
2026-04-02 (Thursday)	Extract EvaluateCareQuality + Shared	B-193	ARCH-001 Phase 4: Extract ClinicalTrialsService + ProviderDetailService	MEDIUM	dev
2026-04-02 (Thursday)	Extract EvaluateCareQuality + Shared	B-194	ARCH-001 Phase 5: Extract Shared Services	MEDIUM	dev
2026-04-03 (Friday)	Pydantic + Cleanup + QA	B-195	ARCH-001 Phase 6: Pydantic tool models	MEDIUM	dev
2026-04-03 (Friday)	Pydantic + Cleanup + QA	B-196	ARCH-001 Phase 7: Delete main.py	LOW	dev
2026-04-03 (Friday)	Pydantic + Cleanup + QA	B-197	Full regression test + SIT	LOW	dev
2026-04-04 (Saturday)	QA Promotion + Stabilization	B-198	Promote to QA	LOW	qa
2026-04-04 (Saturday)	QA Promotion + Stabilization	B-199	Boss UAT on QA	N/A	qa
2026-04-05 (Sunday)	Bug fixes + Production (if needed)	B-200	Fix any UAT bugs	VARIES	qa
2026-04-05 (Sunday)	Bug fixes + Production (if needed)	B-201	Promote to Production	LOW (if QA passed)	prod

Enhancements Included

- F-05: ToolRouter replaces globals() dispatch (Tuesday (Phase 1))
- F-09: Emergency keyword expansion (Tuesday)
- F-10: Provider search cap raised to 25 (Wednesday)
- MAPS: Interactive map for provider search results (Tuesday)

Deferred to Next Sprint

- F-06: Durable Functions status URL auth

- F-08: All-state coverage (pipeline run)
- F-11: FIPS hack resolution (pipeline re-run)
- F-12: DeepSeek privacy disclosure (legal/copy)
- F-13: Lock architecture docs (Boss decision)
- R19: Azure Function timeout UAT
- R35: Paused cluster UAT
- BL-001 through BL-010: Perplexity backlog (v0.1.4)

14. Architectural Decisions

Facade-oriented domains over flat service layer

Two business domains (FindCare, EvaluateCareQuality) each get a facade. Facades are the ONLY entry point from ToolRouter to domain services. Prevents cross-domain coupling.

Pydantic validation at ToolRouter boundary

Claude sends JSON tool arguments. Pydantic validates at the ToolRouter before reaching domain services. Catches schema mismatches early.

System prompt stays in ChatOrchestrator (not PromptBuilder)

Per Boss decision. Prompt references tool names only (not class names). ToolRouter is source of truth for available tools.

Provider Detail belongs to EvaluateCareQuality (not FindCare)

Provider detail is a quality evaluation concern (NPI lookup, board certifications, ratings). Finding a provider and evaluating their quality are distinct business actions.

Frontend remains thin -- no domain duplication

All domain logic stays server-side. Frontend is presentation-only: ChatWindow, MessageBubble, EmergencyBanner. No backend domain structure mirrored.

Strangler Fig migration (not big bang)

7 phases, each deployable. Old functions remain callable until replaced. Zero downtime throughout migration.

No dependency injection for MongoDB/notification clients

Boss rule: e2e tests use real systems, no mocks. DI for testability is unnecessary when you do not mock.

15. Governance Alignment

Policy	Status
GOV-001	No provider prioritization logic in services
GOV-002	Single-tenant config enforced in Config
GOV-004	LLM suggests, ToolRouter enforces execution
GOV-005	Strict three-application separation preserved

16. Sign-Off

Engineer: Claude Opus 4.6

Status: APPROVED

Notes: Design is fully executable. 7-phase migration is incremental and safe. All 18 Perplexity findings addressed.

Architect: GPT-5.3

Status: APPROVED

Notes: Architecture is sound. Facade pattern correctly separates domains. Pydantic enforcement at tool boundary is the right pattern.

Appendix A: AI Model Usage by Prompt

Every AI prompt in the system with the model used, its purpose, and cost profile.

Runtime Prompts (Per User Session)

Prompt	Model	Purpose	Invocation
Chat conversation (user-facing)	Claude Sonnet 4.6 (Anthropic)	Main conversational AI -- tool selection, response generation, intent routing	Every user message
Safety classifier (emergency detection)	GPT-4.1-mini (OpenAI)	Semantic emergency classifier -- second trigger in dual-trigger OR logic	Every user message (parallel with keyword check)
Vector embedding (provider search)	text-embedding-3-large (OpenAI)	Embed user query for vector similarity search against provider descriptions	Every provider search tool call
Vector embedding (specialty search)	text-embedding-3-large (OpenAI)	Embed specialty query for NUCC code matching	Every specialty identification tool call
De-identification (consent flow)	Claude Haiku 4.5 (Anthropic)	Summarize and strip PII from conversation transcript	Only when user declines verbatim but accepts summ consent
Conversation summarization (consent flow)	Claude Haiku 4.5 (Anthropic)	Generate summary of conversation before de-identification	Only when Tier 2 consent granted

Pipeline Prompts (Batch)

Prompt	Model	Purpose	Invocation
Provider description embedding (batch)	text-embedding-3-large (OpenAI)	Generate 3072-dim embeddings for provider vector search index	Pipeline run only (batch, 8.9M records)
County enrichment (geocoding)	N/A (Google Maps API, not AI)	Reverse geocode lat/lng to county FIPS	Pipeline run only

Brain Prompts (Claude + GPT Collaboration)

Prompt	Model	Purpose	Invocation
GPT architecture review / SIT / design collaboration	GPT-5.3-chat-latest (OpenAI)	Enterprise architect role -- design proposals, SIT execution, code audit	On-demand by Claude via brain runner
GPT assurance loop (assignment review)	GPT-5.3-chat-latest (OpenAI)	Review Claude's work against assignments, produce assurance output	Per brain assignment

Model Selection Rationale

Model	Rationale
Claude Sonnet 4.6	Best tool-use performance for conversational AI. Native streaming. Anthropic SDK integration.
GPT-5.3-chat-latest	Reasoning model for architecture and QA. Replaced GPT-4o and o3 after SIT testing showed GPT-4o cannot self-correct in iterative loops and o3 was cost-ineffic (3.3M tokens for 32 iterations). See FIND-SIT-001.
GPT-4.1-mini	Cheapest OpenAI model sufficient for binary safety classification. Called on every message -- cost sensitivity is critical.
Claude Haiku 4.5	Fast, cheap, sufficient for summarization and de-identification. Only invoked during consent flow (rare path).
text-embedding-3-large	3072-dim embeddings for high-quality vector search. OpenAI's best embedding model. Used for both runtime queries and batch pipeline.

End of Document

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